

Subasini K

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Summary

Analytics and Data Science professional with **2+ years** of experience building data-driven decision support systems for operational and business planning. Strong expertise in Python-based analytics, SQL-driven data modeling, forecasting, and insight generation over large-scale transactional datasets. Experienced in translating ambiguous business problems into analytical solutions and deploying scalable AI solutions on cloud platforms.

Skills

- **Programming & Data:** Python, SQL, R, Data Structures
- **Analytics & Modeling:** Statistical Analysis, Time-Series Forecasting, Regression, Classification, Clustering, Model Evaluation
- **Machine Learning:** Tree-based Models, Anomaly Detection, Explainable Models, Large Language Models (LLM), Retrieval-Augmented Generation (RAG), Voice AI
- **Data Engineering:** Pandas, NumPy, Feature Engineering, ETL Pipelines, PySpark (Basics)
- **Cloud & MLOps:** GCP (Cloud Run, Pub/Sub, Vertex AI, Gemini API), AWS (S3, SageMaker, Lambda), Docker, CI/CD
- **Databases & Tools:** MySQL, PostgreSQL, Snowflake, Pinecone, Redis, Git, GitHub, Jupyter, FastAPI, Streamlit, Twilio

Work Experience

Associate AIML Developer – Magilhub

Jul 2024 – Present

- Engineered a production **AI voice ordering system** using **Gemini Live API** and **Twilio** with **Redis**-cached session state for sub-200ms multi-turn conversations; integrated intent recognition and slot-filling logic to handle complex ordering dialogues at scale.
- Designed **RAG pipelines** using the **HyDE (Hypothetical Document Embeddings)** approach for sparse or ambiguous queries – generating a hypothetical answer first and using it as the retrieval vector, significantly improving retrieval accuracy over standard embedding-based search.
- Worked with business, operations, and product stakeholders to translate problems into **AI-driven solutions** and deliver data-backed insights; designed **LLM-based** insight-generation systems converting transactional data into actionable business recommendations.
- Built **Python and SQL ETL pipelines** for data extraction and transformation; developed **forecasting** and anomaly detection logic using historical data to support operational planning and performance evaluation.
- Deployed scalable analytics on **cloud infrastructure** enabling real-time insight delivery through dashboards and APIs for non-technical stakeholders.

AI/ML Intern – Magilhub

Apr 2024 – Jun 2024

- Automated **GenAI-based reporting workflows** on Google Cloud Platform using the **Gemini API**, improving reporting efficiency by **60%**.
- Built data-driven analytics for **operational forecasting** and trend analysis; designed **recommendation models** using collaborative and content-based filtering to improve user engagement.
- Conducted data analysis, experimentation, and **model evaluation** to validate outcomes and ensure business relevance.

Projects

Intelligent Financial Document Analyst

Nov 2025 – Jan 2026

- Architected an end-to-end **RAG system** using **LangChain**, **Pinecone vector database**, and **Claude API** for natural language querying over financial documents – loan agreements, annual reports, insurance policies – with context-aware semantic retrieval.
- Built a **GCP Cloud Run**-hosted **FastAPI** backend with document chunking, embedding generation, and vector ingestion; implemented clause extraction, risk flag detection, and document comparison, reducing review time by **65%**.
- Deployed a **React** frontend on **Vercel** with real-time query interface serving retail investors and finance professionals.

Explainable ML Credit Scoring System

Feb 2026 – Apr 2026

- Trained an **XGBoost** credit risk model on the **LendingClub** dataset with feature engineering and cross-validation, achieving **91% AUC-ROC** and **88% F1-score**; integrated **SHAP** for per-decision plain-language explanations satisfying **Explainable AI (XAI)** standards.
- Deployed on **HuggingFace Spaces** with an interactive **Streamlit** UI featuring real-time score prediction and SHAP waterfall charts.

Empathetic Response AI – Conversational RAG Chatbot

Aug 2025 – Sep 2025

- Built a production **RAG conversational AI chatbot** using **MCP (Model Context Protocol)** and **Vertex AI**, supporting multi-turn natural language queries over operational databases via **vector search** and **LLM reasoning**; reduced manual data lookup time by **50%**.
- Implemented retrieval quality evaluation using cosine similarity scoring and human feedback loops, reducing hallucination rates by **40%**.

Multimodal Fingerprint Liveness Detection

Nov 2023 – May 2024

- Designed a **multimodal deep learning pipeline** combining texture and frequency-domain **feature fusion** with **OpenCV** preprocessing (Gaussian blur, histogram equalization, CLAHE) to classify live fingers vs. spoofs; trained a **SVC** model improving detection accuracy by **98%** and reducing **FAR** from 12% to under 1%.

- Applied multi-scale image augmentation and cross-sensor generalization across optical, capacitive, and ultrasonic fingerprint scanners.

Education

B.Tech in Artificial Intelligence & Data Science

2020 – 2024

Mepco Schlenk Engineering College, Sivakasi, TN

CGPA: 7.85 / 10.0

Certifications & Awards

- **AWS Certified AI Practitioner** – Amazon Web Services (AWS), 2026
- **Innovation Award** – Q1 & Q2 2025, Magilhub. [View Certificate](#)
- **Trainee Excellence Award** – Q1 2024, Magilhub. [View Certificate](#)